

PAUL MITCHELL

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SOFTWARE ENGINEERING · CLOUD SYSTEMS & DISTRIBUTED PLATFORMS

PROFESSIONAL SUMMARY

William & Mary Computer Science M.S. candidate (3.95 GPA, Systems & AI/ML concentration) and B.S. graduate targeting Capital One's Technology Development Program (TDP). Combines production-grade Linux systems administration (100+ servers), event-driven microservice development, and AWS cloud pipelines with hands-on experience building ultra-low-latency API middleware (0.038ms P99) and distributed telemetry architectures. Adept in high-throughput data streaming, resilient systems design, and test-driven deployment.

EDUCATION

College of William & Mary — M.S. in Computer Science (AI/ML & Systems) Expected Dec 2026 · GPA: 3.95
College of William & Mary — B.S. in Computer Science Graduated Dec 2024
Relevant Coursework: Machine Learning Systems, Advanced Software Engineering, Deep Learning, Distributed Systems, Data Mining

TECHNICAL SKILLS & INFRASTRUCTURE

Cloud & Distributed Systems: AWS (EC2, S3, SQS, Lambda, IAM), Docker, Apache Kafka, Redis, microservices, Linux (Ubuntu/RHEL), Bash
Programming & Back-End: Python, FastAPI, Java, Go, C/C++, SQL (PostgreSQL), Bash, Git/GitLab CI, RESTful APIs, TDD, OOP
Observability & Reliability: Prometheus, Grafana, OpenTelemetry, metrics instrumentation, alert rule engineering, incident triage
Data & Machine Learning: PyTorch, scikit-learn, LightGBM, DistilBERT, Pandas, NumPy, stream processing, anomaly detection

TECHNICAL & PROFESSIONAL EXPERIENCE

IT Administrator | William & Mary Technology Support Center | Williamsburg, VA Feb 2025 – Present

- Administer and maintain 100+ Linux instructional compute servers, orchestrating automated system configurations via Puppet, DNS management, and network troubleshooting across multi-room labs.
- Architected Prometheus and Grafana telemetry pipelines monitoring cluster uptime, storage utilization, and compute health, cutting incident detection and response times by over 40%.
- Evaluated multi-user Docker containerization architectures, producing technical documentation on isolation boundaries, resource limits, and operational maintainability for department heads.
- Standardized network routing and static IP configurations, maintaining strict configuration control across all production instructional hosts.
- Developed automated Bash recovery scripts for stalled network daemons, eliminating recurring manual administrator interventions.

Data Analytics & Engineering Extern | Extern, Amazon-Sponsored Project | Remote Jun 2025 – Sep 2025

- Engineered reproducible automated Python and AWS S3 data pipelines to clean, validate, and structure ~20,000 text records, optimizing data prep execution cycles from hours to minutes.
- Developed a predictive classification model using LightGBM and deployed an interactive Streamlit analytics platform delivering real-time decision-support metrics for program mentors.
- Implemented cloud dataset versioning using DVC on AWS S3 to ensure reproducibility and audit compliance across multi-team evaluations.
- Fine-tuned DistilBERT transformer models (F1 = 0.89) to categorize multi-topic survey corpora into prioritized operational action items.

KEY TECHNICAL & SYSTEMS PROJECTS

ChaosLedger: Zero-Discrepancy Stream Engine under Simulated AWS Partitioning | Python, Kafka, Redis 2026

- Architected a distributed financial stream engine simulating 100k tx/min under chaos-injected 200ms latency spikes and worker failures; guaranteed exactly \$0.00 balance discrepancy using atomic Redis idempotency.
- Engineered adaptive consumer-lag backpressure preventing Kafka partition rebalance storms during network splits; held P99 latency strictly under 6ms with Prometheus/Grafana real-time telemetry.

ShadowPrompt: In-Memory Pre-Inference Defense Proxy | Python, FastAPI, Docker 2026 · [pjmitchell7.github.io/shadowprompt](https://github.com/pjmitchell7/shadowprompt)

- Engineered an asynchronous reverse proxy evaluating incoming HTTP streams in memory at 0.038ms P99 latency, neutralizing steganographic injection and payload assembly attacks prior to model ingestion.
- Implemented zero-copy token boundary scanners eliminating redundant serialization overhead, achieving 25,000+ req/sec throughput.

PromptShield: Tokenization-Aware LLM Security Middleware | Python, FastAPI, scikit-learn 2026

- Built asynchronous FastAPI middleware intercepting adversarial prompt injections with 0.27–1.1ms latency, avoiding expensive secondary LLM calls.
- Enforced strict probability thresholds keeping false positive rates under 0.1% across complex banking queries and schema validation payloads.